FIGHTER UAV COMPETITION

CRITICAL DESIGN REPORT TEMPLATE

TEAM NAME:

TEAM ID:

APPLICATION ID:

2026

TEKNOFEST

AEROSPACE AND TECHNOLOGY FESTIVAL

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1. BASE SYSTEM SUMMARY (5 Points)

1.1 System Description

In this section, the system's mission description and features are briefly explained. The definitions of the components that make up the general system (Ground Control Station, Ground Antenna, Remote Control, Aircraft etc.) and their functions within the system are also mentioned.

1.2 System Final Perfomance Specifications

In this section should include the finalized aircraft performance values. (e.g. Wingspan, cruise speed, take-off weight, stall speed, communication range, flight time, etc.). The performance values to be included here should be shown in tabular form.

2. ORGANIZATION SUMMARY (4 Points)

2.1 Team Organization

In this section, the organizational chart of the team and the duties of the units are briefly mentioned. (Personal information of team members such as names and photographs SHOULD NOT be included.)

2.2 Timeline and Budget

In this section, the estimated budget and actual budget differences should also be included, as well as differences between the planned timeline and the actual flow chart. (delays, early completion, etc.)

3. DETAILED DESIGN SUMMARY (15 Points)

3.1 3D Design of Aircraft

In this chapter; The dimensions of the aircraft, the in-vehicle placement of the subsystems, the location of the fuse placed within the scope of security needs should be included in the visuals. At the same time, during the technical control on the time of the competition, the designed vehicle and the vehicle arriving at the site will be compared with the visuals given here.

3.2 Aircraft Performans Summary

In this section, the flight time calculations that the aircraft can stay in the air during a competition round should be shown in detail.

In addition, the analyzes performed (e.g. aerodynamic analysis, structural analysis, etc.) should be explained in this section.

3.3 Final System Architecture and Subsystems Summary

In this section, the final system architecture should include the brand/model information of the hardware to be used. It is expected to show all power and communication lines in the system architecture. It is requested to add the system architecture visually.

In addition to the system architecture, the technical specifications of the selected subsystems and their compatibility with each other should be explained. If there are multiple options investigated for the same task, it should be explained why the final product was chosen. This section should explain how the frequency bands for the RF devices used are selected.

3.4 Aircraft Weight Distribution

In this section, the weight distribution of the aircraft at the subsystem level will be given. (All sub-components' Weights' and positions' according to chosen reference point should be given. Additionally, center of gravity of the air vehicle must be given according to wing tip.)

4. AUTONOMOUS MISSIONS (25 Points)

4.1 Autonomous Lockdown

In this section, it should be explained how the team will evaluate the data received from the server and decide on the most suitable opponent aircraft to track. It should be explained how to approach the aircraft deemed suitable for tracking and how to proceed during the approach.

The posture controls (lying down, standing up, turning) to be used to keep the QR code image within the Hit Target Area during the kamikaze maneuver must be explained in detail. How the object detection and tracking algorithms to be used are selected/developed should be explained. In addition to this explanation, it should be compared with alternative methods and their advantages/disadvantages should be mentioned.

In the designed algorithm, information such as how the aircraft will control the posture (aircraft's tilt, upwards-downward and orientation angles) and speed in response to the error in the image should be given. In addition to this explanation, it should be compared with alternative methods and their advantages/disadvantages should be explained.

The algorithms described in this section can be supported with flow diagrams, controller diagrams, graphs or figures.

4.2 Kamikaze Mission

In this section, how the UAV will perform the Kamikaze mission and the phases of approaching the target (approach), diving and passing phases should be explained in detail (e.g. dive angle, wind direction during approach, diving and passing altitude, maximum speed to be reached during diving, etc.).

The tilt, upwards-downward controllers to be used during kamikaze should be explained in detail. The structural strength of the aircraft during the mission (e.g. the maximum G force that the aircraft will be exposed to during diving and passing, etc.) should be explained in detail.

It should be explained how the QR target will be detected with an algorithm and how the QR data obtained will be transferred to the server. It should be explained how to prevent distortions that may occur in the image, and from how many meters altitude the camera to be used can read the QR target.

The algorithms described in this section can be supported with flow diagrams, controller diagrams, graphs or figures.

5. AIR DEFENSE SYSTEM (5 Points)

This section should detail how to avoid restricted areas when the air defense system is active. Autonomous avoidance or route planning algorithms, if used, should be described. If the avoidance is not autonomous, the avoidance steps to be performed by the operator should be detailed.

The algorithms described in this chapter can be supported by flow diagrams, controller diagrams, graphs or figures.

6. GROUND STATION, COMMUNICATION AND USER INTERFACE DESIGN (12 Points)

In this section, the communication between the aircraft and the ground control station should be explained in detail (e.g. antenna frequencies, communication protocols, communication hardware, image transmission systems, antenna tracker, modem, router, communication with the contest server, etc.).

In addition, the design(s) of the interface(s) to be used in the ground control station will be explained. Here, the location of information such as speed, altitude, mode change, locking quadrilateral, air defense systems, flight limits, instant visual displays of rival aircraft, if any, should be shown with visuals on the interface/interfaces.

(Actual boundaries will be shared with the competitors on the day of the competition. The boundaries to be shown on the CDR can be determined hypothetically by the teams).

7. AIRCRAFT INTEGRATION (10 Points)

7.1 Structural and Mechanical Integration

In this section, it should be explained how the aircraft is manufactured/assembled, how structural integrity is ensured, how assembly is carried out, how moving parts such as control surface/engine are fixed and supported with visuals (e.g. wing, fuselage, tail, landing gear, engine, etc.).

7.2 Electronic Integration

In this section, the placement of the electronic components (sensors, avionics, actuators, camera, battery, etc.) inside the aircraft, how they are fixed, what type of cables/connectors are used for in-vehicle wiring, ground/fuse connections applied inside the aircraft should be explained and the steps performed should be supported with visuals.

8. TEST AND SIMULATION (15 Points)

8.1 Subsystem Tests

This section will contain the tests that are planned to be done during vehicle development. If any, the simulation systems developed and the tests performed on them should also be included here. Each test performed should be included as a subtitle in this section. For example: 8.1.2 Thrust Test, 8.1.3 Structural Test, 8.1.4 Communication Range Test...

8.2 Flight Test and Flight Checklist

This section should include the 'Flight Checklist' of the system before the flight.

At the same time, visuals of the flight tests, if any, and the results of these tests should be shared in a compiled form. If no flight tests have been conducted yet, the number of flight tests planned to be conducted until the date of the competition, test objectives and expected results should be given (e.g. maximum flight time test, autonomous landing test, etc.).

8.3 Mission Tests

Visuals of the details of the competition tasks performed in flight tests and the results of these tests should be shared in a compiled form (e.g. Dive Test, Lock-on Test, Evading Air Defense System, etc.). Support should be obtained from simulation environments to test the algorithms planned to be used in the competition. The aforementioned tests can be performed as software-in-the-loop (SITL) or hardware-in-the-loop (HITL). Various simulation environments can be used to simulate the world environment mathematically or visually (e.g. Flightgear, XPlane, Gazebo, Unity, Unreal Engine). The methods, implementation steps and results of the tests should be shared.

9. SAFETY (5 Points)

In this section, possible risk and mitigation for this risks should be explained. (Ex: first aid kit on working area, using special battery pack, etc.)

10. REFERENCES

This section should include references.

NOTES:

All reports should be written in accordance with academic report standards.

- Information about the contents of the reports are stated above.

- All reports should contain "Table of Contents" and "References".

- Each report should include a cover page.

- Reports pages should be numbered consecutively.

-All pages expect the cover page should contain in the footer the team's name and page number, in the header "Fighter UAV Competition 2024".

- Font: Arial, Size: 12, Line Spacing: 1.15

-All the paragraphs should be aligned to both sides (Justify).

- Page Layout: A4 Type and Margins 2.5 cm

- The report should not exceed 25 pages.

- Our teams that have benefited from the reports of the past years on our website have to include cited phrase after the sentence cited.

CITED FORMAT: "Cited Phrase/s (Year, Competition Name, Category, Team Name)
EXAMPLE QUOTE: "Failure to determine the location of the earthquake victim in the wreckage is the most important problem that slows down the debris removal and search for the earthquake victim." (2022, Technology For Humanity, Disaster Management, X Team)

Critical Design Report scoring will be based on the template below.

Part		Scoring
1	Basic System Summary	5
2	Organization Summary	4
3	Detailed Design Summary	15
4	Autonomous Missions	25
5	Air Defence System	5
6	Ground Station and Communication, User Interface Design	12
7	Aircraft Integration	10
8	Test and Simulation	15
9	Safety	5
10	Report Layout	4